

# Onyx — PrinterKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **PrinterKit** (`SD:/Lib/printerkit.so`, `user/Kits/printerkit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

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2. Using it
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4. `printerkit/printerkit.h`

## What it is

PrinterKit is printing: the Print dialog, a job and its pages — text, lines, shapes, pictures — handed to the print service, which makes of them what the printer takes, or a PDF file.

|         |                                                                                                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|
|         |                                                                                                                                   |
| Include | <code>#include "printerkit/printerkit.h"</code>                                                                                   |
| Link    | <code>lib/printerkit.imp.a</code>                                                                                                 |
| Library | <code>SD:/lib/printerkit.so</code> — 33 entries in its table<br>( <code>user/Kits/printerkit/printerkit.abi</code> , append-only) |
| Sources | <code>user/Kits/printerkit/</code>                                                                                                |

## Using it

A program prints in four steps: the Print dialog, a job, its pages, the end. Lengths are points (1/72 inch) from the page's top-left corner; colours are `0xRRGGBB`. The pages are recorded: the print service turns them into what the printer takes — or into a PDF file.

```
#include "printerkit/printerkit.h"
```

```
PrintSetup s;  
print_setup_default (&s);  
PrintDialogInfo di = { sizeof di, "My document", 1, 0, 0, 0, 0 };  
if (print_dialog (&s, &di)) // the printer, the paper, the pages  
{  
    PrintJob *j = print_begin (&s, "My document");  
    int font = print_font (j, "DejaVu Sans", 0);  
    if (print_page (j, 0, 0)) // 0, 0: the paper chosen  
    {  
        print_text (j, font, 12, 72, 84, "Hello, printer", 0x000000);  
        print_line (j, 72, 90, s.paper_w - 72, 90, 0.75f, 0x808080);  
    }  
}
```

```
print_end (j); // to the print queue
}
```

Rectangles, paths (lines and curves, filled or stroked) and pictures are drawn the same way (`print_rect`, `print_path_*`, `print_image`).

## Index

Everything the headers declare, in their order — the details are in each header's part below.

| Name                             | What it does                                                 | Header                    |
|----------------------------------|--------------------------------------------------------------|---------------------------|
| <code>PrintSetup</code>          | What the user chose in the Print dialog.                     | <code>printerkit.h</code> |
| <code>PrintDialogInfo</code>     | the page is fitted on the printer's paper                    | <code>printerkit.h</code> |
| <code>print_setup_default</code> | the default printer, its default paper, upright, one copy    | <code>printerkit.h</code> |
| <code>print_dialog</code>        | 1: print (s filled), 0: cancelled                            | <code>printerkit.h</code> |
| <code>print_setup_paper</code>   | another paper -> 1, 0 unknown name                           | <code>printerkit.h</code> |
| <code>print_begin</code>         | 0: the spool cannot be written                               | <code>printerkit.h</code> |
| <code>print_page</code>          | a new page (w, h 0: the setup's paper) -> 1, or 0: outside   | <code>printerkit.h</code> |
| <code>print_rect</code>          | the pages to print -- skip its drawing (it would be ignored) | <code>printerkit.h</code> |
| <code>print_line</code>          |                                                              | <code>printerkit.h</code> |
| <code>print_frame</code>         | a rectangle's outline                                        | <code>printerkit.h</code> |
| <code>print_path_move</code>     | a shape                                                      | <code>printerkit.h</code> |
| <code>print_path_line</code>     |                                                              | <code>printerkit.h</code> |
| <code>print_path_curve</code>    |                                                              | <code>printerkit.h</code> |
| <code>print_path_close</code>    |                                                              | <code>printerkit.h</code> |
| <code>print_path_fill</code>     |                                                              | <code>printerkit.h</code> |
| <code>print_path_stroke</code>   |                                                              | <code>printerkit.h</code> |
| <code>print_image</code>         | an image                                                     | <code>printerkit.h</code> |
| <code>print_font</code>          | text                                                         | <code>printerkit.h</code> |
| <code>print_text</code>          |                                                              | <code>printerkit.h</code> |
| <code>print_text_width</code>    |                                                              | <code>printerkit.h</code> |
| <code>print_font_metrics</code>  |                                                              | <code>printerkit.h</code> |

| Name                  | What it does                                                | Header       |
|-----------------------|-------------------------------------------------------------|--------------|
| print_font_data       | for an app with its own text layout                         | printerkit.h |
| print_glyph           |                                                             | printerkit.h |
| print_end             | the job queued -> its number (> 0), or < 0: it could not be | printerkit.h |
| print_abort           | nothing printed                                             | printerkit.h |
| PrintPrinter          | (a type)                                                    | printerkit.h |
| print_printers        | -> how many                                                 | printerkit.h |
| print_printer_media   | a paper of a printer                                        | printerkit.h |
| print_printer_add     | 1 / 0 (err: why)                                            | printerkit.h |
| print_printers_find   |                                                             | printerkit.h |
| print_printer_remove  |                                                             | printerkit.h |
| print_printer_default |                                                             | printerkit.h |
| print_printer_status  | asked now: 1 ready (text: ink...), 0 (text: why)            | printerkit.h |
| print_jobs            | -> how many (the queue, then the last finished)             | printerkit.h |
| print_job_cancel      |                                                             | printerkit.h |
| print_jobs_forget     | the finished jobs dropped from the list                     | printerkit.h |

### printerkit/printerkit.h

printerkit/printerkit.h -- printing from an app (the shared library SD:/lib/printerkit.so; docs/03 "Printing").

An app prints in four steps -- the Print dialog, a job, its pages, the end:

```

PrintSetup s; print_setup_default (&s);
PrintDialogInfo di = { sizeof di, "My document", pageCount, 0, 0, 0, 0 };
if (print_dialog (&s, &di)) // the user chose a printer, a paper,
the pages...
{
    PrintJob *j = print_begin (&s, "My document");
    int f = print_font (j, "DejaVu Sans", 0);
    for (int p = 0; p < pageCount; p++)
        if (print_page (j, 0, 0)) // 0, 0: the paper chosen; 0
returned: not in the range
        {
            print_text (j, f, 12, 72, 72 + 12, "Hello", 0x000000);
            print_line (j, 72, 90, s.paper_w - 72, 90, 0.75f, 0x808080);
        }
    print_end (j); // the job goes to printd's queue
}

```

Lengths are points (1/72 inch), y goes down from the page's top-left corner, colours are 0xRRGGBB. The pages are recorded, not rendered: the print service (printd) makes of them what the printer takes -- a PDF file for the PDF printer, pages of pixels at its resolution for a network printer -- and tells the user when the job is done. The app does not wait for the printer.

A program links lib/printerkit.imp.a (with lib/uikit.imp.a and lib/fontkit.imp.a: the dialog is uikit's, the text FreeType's). The table is append-only (printerkit/printerkit.abi): these functions never change.

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```
typedef struct PrintJob PrintJob;

enum { PRINT_DRAFT = 3, PRINT_NORMAL = 4, PRINT_HIGH = 5 };    // quality
enum { PRINT_PORTRAIT = 0, PRINT_LANDSCAPE = 1 };
enum { PRINT_BOLD = 1, PRINT_ITALIC = 2 };                    // a font's style
enum { PRINT_IMG_ALPHA = 1,    // the pixels are 0xAARRGGBB (else 0x00RRGGBB: opaque)
      PRINT_IMG_PHOTO = 2 };    // a photo: kept as a JPEG (smaller; not for text or line
art)
```

What the user chose in the Print dialog. Fields are only ever added at the end (size says how many).

```
typedef struct PrintSetup
{
    unsigned size;    // sizeof (PrintSetup)
    char printer[64];    // the printer's name ("PDF": the PDF printer)
    char media[64];    // the paper, as IPP names it ("iso_a4_210x297mm")
    float paper_w, paper_h;    // the page in points, the orientation applied
    float margin_l, margin_t, margin_r, margin_b;    // what the printer cannot print on,
along each edge
    int orientation;    // PRINT_PORTRAIT / PRINT_LANDSCAPE
    int copies;
    int mono;    // 1: black and white
    int quality;    // PRINT_DRAFT / NORMAL / HIGH
    int from, to;    // the pages to print, from 1; 0, 0: all
    char output[256];    // the PDF printer: the file to write
    int reserved[16];
} PrintSetup;
```

```
enum { PRINT_DLG_OWN_PAPER = 1 };    // the document has its own page size (a slide, a PDF):
no paper choice --
```

the page is fitted on the printer's paper

```
typedef struct PrintDialogInfo
{
    unsigned size;    // sizeof (PrintDialogInfo)
    const char *title;    // the document's name (the dialog's title, the PDF's file
name)
    int pages;    // how many pages it has (0: not known)
    int current;    // the page shown (from 1: offers "Current page"; 0: no such
```

```
choice)
    int flags; // PRINT_DLG_*
    float paper_w, paper_h; // the document's own paper to select first (0: the
printer's default)
} PrintDialogInfo;
```

## the dialog

```
void print_setup_default (PrintSetup *s); // the default printer, its default
paper, upright, one copy
int print_dialog (PrintSetup *s, const PrintDialogInfo *info); // 1: print (s filled), 0:
cancelled
int print_setup_paper (PrintSetup *s, const char *media, int orientation); // another paper
-> 1, 0 unknown name
```

## a job

```
PrintJob *print_begin (const PrintSetup *s, const char *title); // 0: the spool cannot be
written
int print_page (PrintJob *j, float w, float h); // a new page (w, h 0: the setup's paper)
-> 1, or 0: outside
```

the pages to print -- skip its drawing (it would be ignored)

```
void print_rect (PrintJob *j, float x, float y, float w, float h, unsigned rgb);
void print_line (PrintJob *j, float x0, float y0, float x1, float y1, float width, unsigned
rgb);
void print_frame (PrintJob *j, float x, float y, float w, float h, float width, unsigned
rgb); // a rectangle's outline
```

a shape: move to, lines and Bezier curves, closed or not; then filled, or stroked with a line

```
void print_path_move (PrintJob *j, float x, float y);
void print_path_line (PrintJob *j, float x, float y);
void print_path_curve (PrintJob *j, float x1, float y1, float x2, float y2, float x3, float
y3);
void print_path_close (PrintJob *j);
void print_path_fill (PrintJob *j, unsigned rgb);
void print_path_stroke (PrintJob *j, float width, unsigned rgb);
```

an image: pw x ph pixels drawn in (x, y, w, h)

```
void print_image (PrintJob *j, const unsigned *px, int pw, int ph, float x, float y, float
w, float h, int flags);
```

text: a font by its family ("DejaVu Sans", "Liberation Serif"...: SD:/res/fonts, SD:/fonts) and style -> its number (-1: no such font nor a fallback); the text at its baseline -> the pen's advance

```
int print_font (PrintJob *j, const char *family, int style);
float print_text (PrintJob *j, int font, float size, float x, float baseline, const char
*utf8, unsigned rgb);
float print_text_width (PrintJob *j, int font, float size, const char *utf8);
void print_font_metrics (PrintJob *j, int font, float size, float *ascent, float *descent,
float *line);
```

for an app with its own text layout: a TrueType file's bytes (kept by the app until print\_end) -> the font's number; one glyph of it at its pen position (style: PRINT\_BOLD / ITALIC made from this face)

```
int print_font_data (PrintJob *j, const unsigned char *ttf, unsigned len);
void print_glyph (PrintJob *j, int font, float size, float x, float baseline, unsigned
glyph, unsigned unicode, unsigned rgb, int style);
```

```
int print_end (PrintJob *j); // the job queued -> its number (> 0), or < 0: it could
not be
void print_abort (PrintJob *j); // nothing printed
```

## the printers and the queue (the Control Panel's applet; an app's own status)

```
typedef struct PrintPrinter
{
    char name[64], kind[8], uri[160], model[64]; // kind: "pdf", "ipp"
    int color, copies, dpi, is_default;
    unsigned quality; // bits 3 (draft), 4 (normal), 5 (high)
    char media_default[64];
} PrintPrinter;
enum { PRINT_JOB_QUEUED = 0, PRINT_JOB_PREPARING, PRINT_JOB_SENDING, PRINT_JOB_PRINTING,
PRINT_JOB_DONE, PRINT_JOB_FAILED, PRINT_JOB_CANCELED };
typedef struct PrintJobInfo { unsigned id; int state, pages, page; char printer[64],
title[80], message[96]; } PrintJobInfo;
```

```
int print_printers (PrintPrinter *out, int max); // -> how many
```

a paper of a printer: its IPP name, its name for a person -> 1, 0: no i-th paper

```
int print_printer_media (const char *printer, int i, char *name, int ncap, char *label, int
lcap);
```

a network printer asked what it can do, and added (address: "192.168.0.14", "ipp://host:631/ipp/print")

```
int print_printer_add (const char *name, const char *address, char *err, int cap); // 1 / 0
(err: why)
```

the local network searched for printers (mDNS / DNS-SD, as AirPrint: a few seconds) -> how many;  
usable 0: it answers but takes neither PWG Raster nor PDF

```
typedef struct PrintFound { char address[32], name[64], model[64]; int usable; } PrintFound;
int print_printers_find (PrintFound *out, int max);
int print_printer_remove (const char *name);
int print_printer_default (const char *name);
int print_printer_status (const char *name, char *text, int cap); // asked now: 1 ready
(text: ink...), 0 (text: why)
int print_jobs (PrintJobInfo *out, int max); // -> how many (the queue, then the
last finished)
int print_job_cancel (unsigned id);
void print_jobs_forget (void); // the finished jobs dropped from the list
}
#endif
```